



Artificial Intelligence Foundations (CIS4049-N)

Week 2 – Intelligent Agents – Part 3

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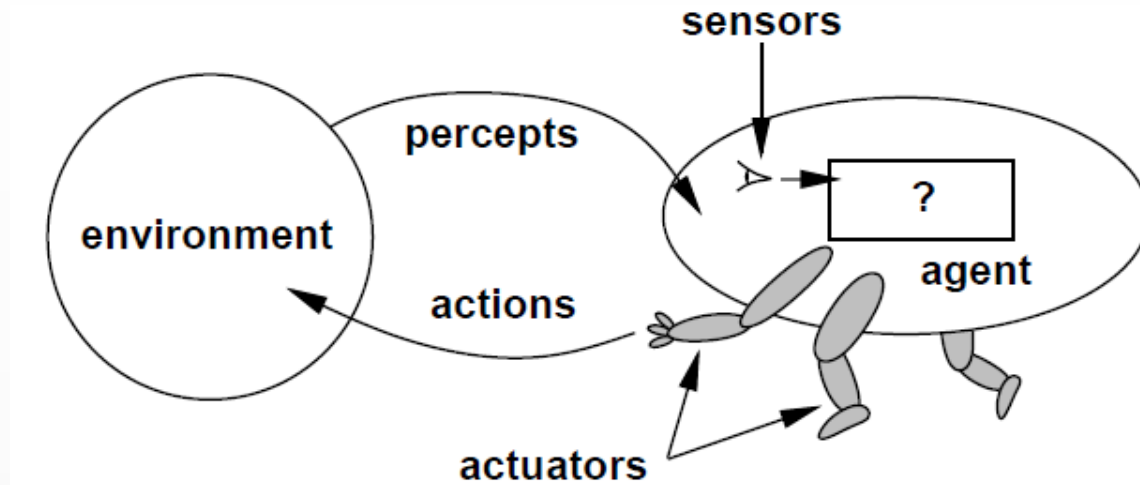
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Designing Rational Agents

Designing Rational Agents



Agents

- **agent functions** map sequences of **observations** to **actions**:

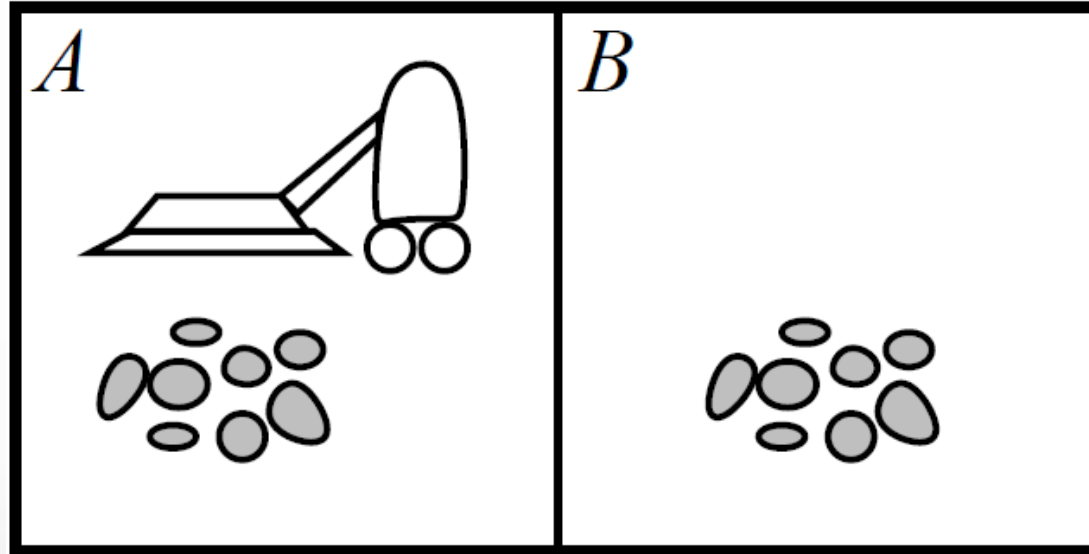
$$f : \mathcal{P}^+ \rightarrow \mathcal{A}$$

- **agent program**: runs on physical **architecture** and computes f

Introducing: an Agent



Vacuum domain



- **Observations:** location and cleanness of current room:
 $\langle a; \text{clean} \rangle$, $\langle a; \text{dirty} \rangle$, $\langle b; \text{clean} \rangle$, $\langle b; \text{dirty} \rangle$
- **Actions:** left, right, suck, wait

Vacuum Agent

- A possible *agent function*:

observation sequence	action
$\langle a, \text{clean} \rangle$	<i>right</i>
$\langle a, \text{dirty} \rangle$	<i>suck</i>
$\langle b, \text{clean} \rangle$	<i>left</i>
$\langle b, \text{dirty} \rangle$	<i>suck</i>
$\langle a, \text{clean} \rangle, \langle b, \text{clean} \rangle$	<i>left</i>
$\langle a, \text{clean} \rangle, \langle b, \text{dirty} \rangle$	<i>suck</i>
...	...

Reflexive (or Reflex) Agents

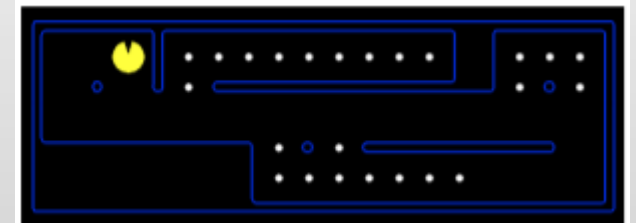
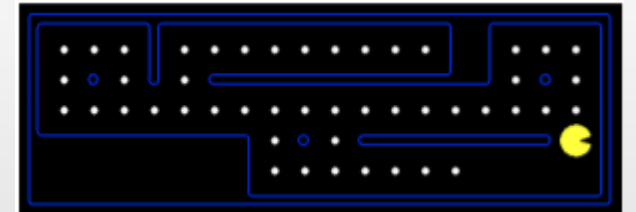
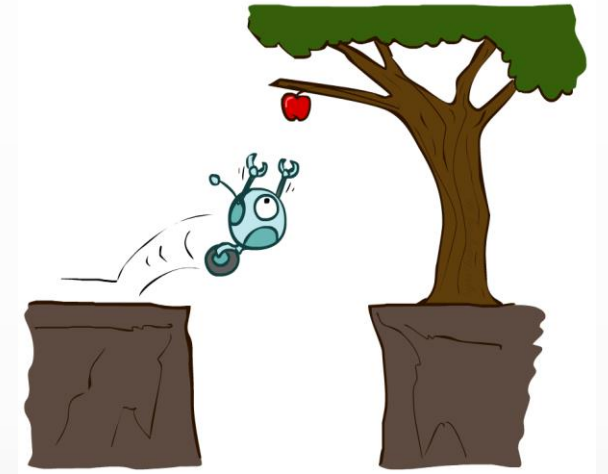
- **Reflexive agents** compute next action only based on **last observation** in sequence:
 - very **simple model**
 - very **restricted**
 - corresponds to Mealy automaton (a kind of DFA)
 - with only **1 state**

Example (A Reflexive Vacuum Agent)

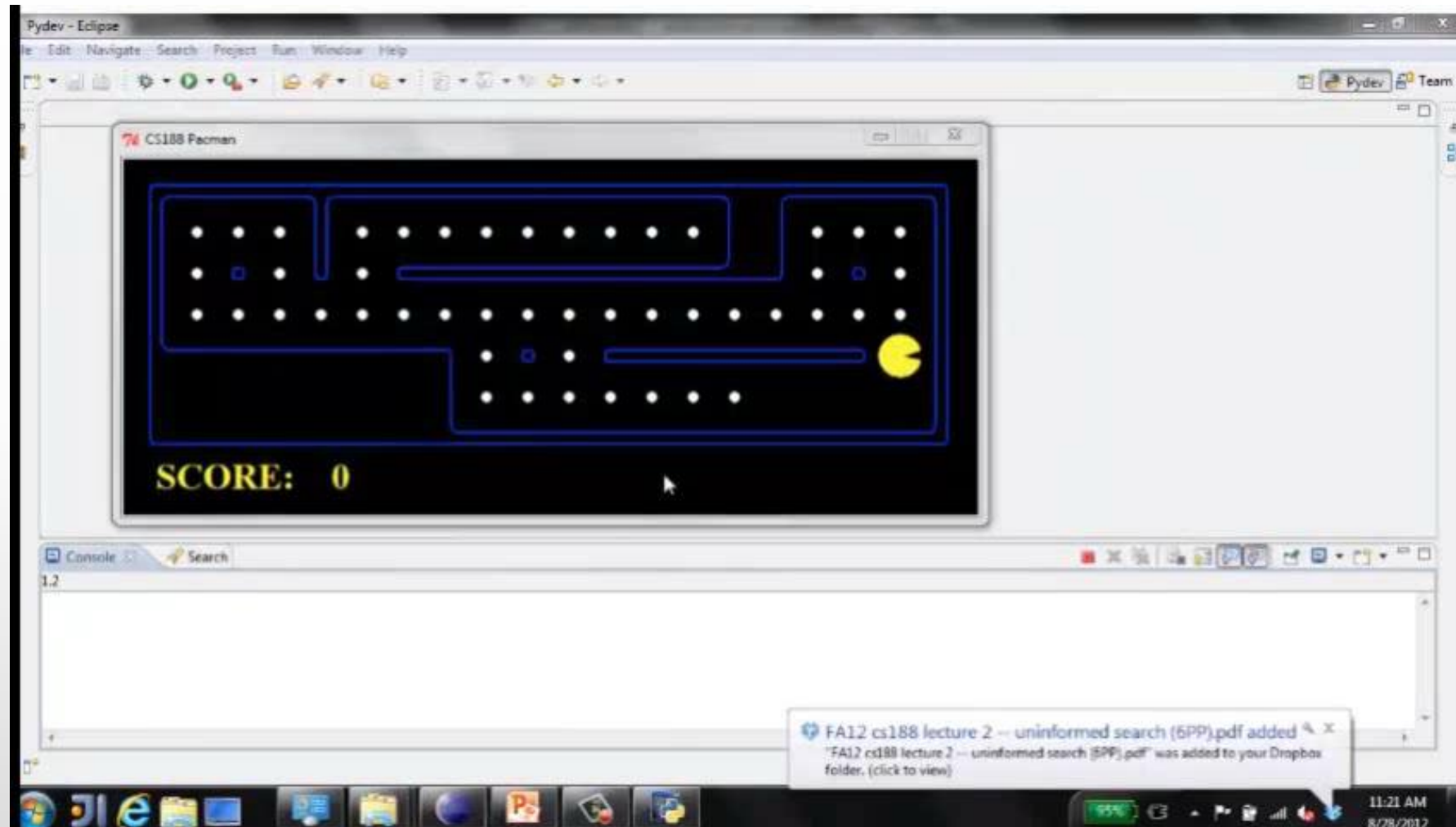
```
def reflex-vacuum-agent(location, status):  
    if status = dirty: return suck  
    else if location = a: return right  
    else if location = b: return left
```

Reflex Agents

- Choose **action** based on **current percept** (and maybe memory)
- May have memory or a model of the world's current state
- **Do not consider** the **future consequences** of their **actions**
- Consider how the world IS.
- Can a reflex agent be rational?
- Examples:
 - Stove: hot, ouch!
 - Car: deer, brake!
 - Go to Tahoe: skid in snow, ???



Video of Demo Reflex Odd



How to design rational Agents?

- Evaluating agent functions - What is the right **agent function**?

Rational Behavior

Evaluate behavior of agents with **performance measure** (related terms: **utility**, **cost**).

perfect rationality:

- always select an action **maximizing**
- **expected value** of **future performance**
- given **available information** (observations so far)

Is our Agent perfectly rational?

- **Question:** Is the reflex vacuum agent of the example **perfectly rational**?
- Depends on **performance measure** and **environment**!
 - Do actions reliably have the desired effect?
 - Do we know the initial situation?
 - Can new dirt be produced while the agent is acting?

Rational Vacuum Agent

Example (Vacuum Agent)

performance measure:

- +100 units for each cleaned cell
- -10 units for each *suck* action
- -1 units for each *left/right* action

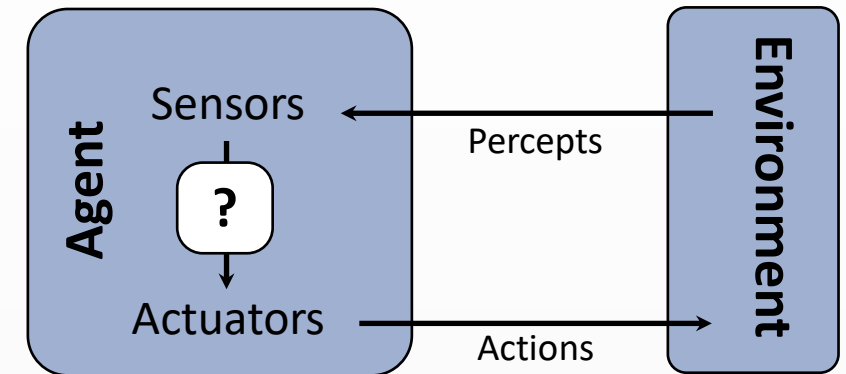
environment:

- actions and observations reliable
- world only changes through actions of the agent
- all initial situations equally probable

How should a perfect agent behave?

Rationality

- A **rational agent** chooses actions that **maximize expected utility**
- Today: agents that have a goal, and a cost - e.g., reach goal with lowest cost
- Later: agents have numerical utilities, rewards, etc. e.g., takes action that maximizes total reward over time. (reward: largest total profit. or expected total profit.)



Rationality

- What is rational at any given time depends on four things:
 - The **performance measure** that defines the criterion of success.
 - The agent's **prior knowledge** of the **environment**.
 - The **actions** that the agent can perform.
 - The **agent's percept sequence** to date.
- Defining a **rational agent**:

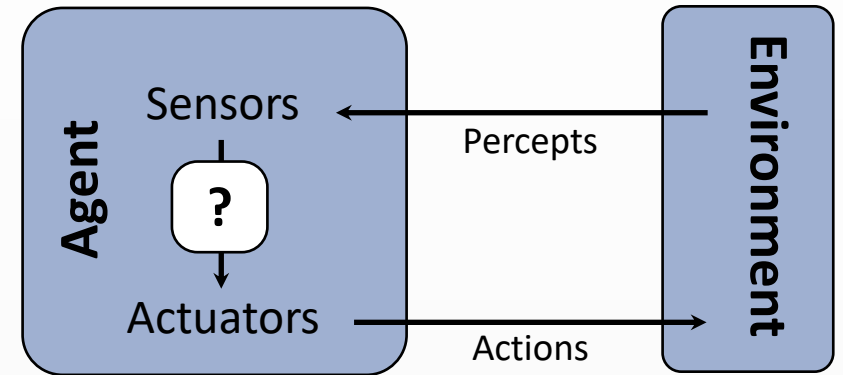
*“For each possible percept sequence, a rational agent should select **an action** that is expected to **maximize its performance measure**, given the evidence provided by the **percept sequence** and **whatever built-in knowledge** the agent has.”*

Rationality - Discussion

- Perfect rationality $\neq$ omniscience
 - Incomplete information (due to limited observations) reduces achievable utility
- Perfect rationality $\neq$ perfect prediction of future
 - Uncertain behavior of environment (e.g., stochastic action effects) reduces achievable utility
- Perfect rationality is rarely achievable
 - Limited computational power, bounded rationality

Summary

- Common **metaphor** for **AI systems**: **rational agents**
- “An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through effectors” (*Artificial Intelligence: A Modern Approach*” by S. Russell and P. Norvig)
- **Agent interacts with environment**: sensors **perceive** observations about state of the environment, and **actuators** perform actions modifying the **environment**
- Formally: agent function maps **observation sequences** to **actions**
- **Reflexive agent**: agent function only based on last observation



Summary

- **Rational agents:** try to maximise performance measure (utility)
- **Perfect rationality:** achieve maximal utility in expectation given available information
- For “interesting” problems rarely achievable: **bounded rationality**